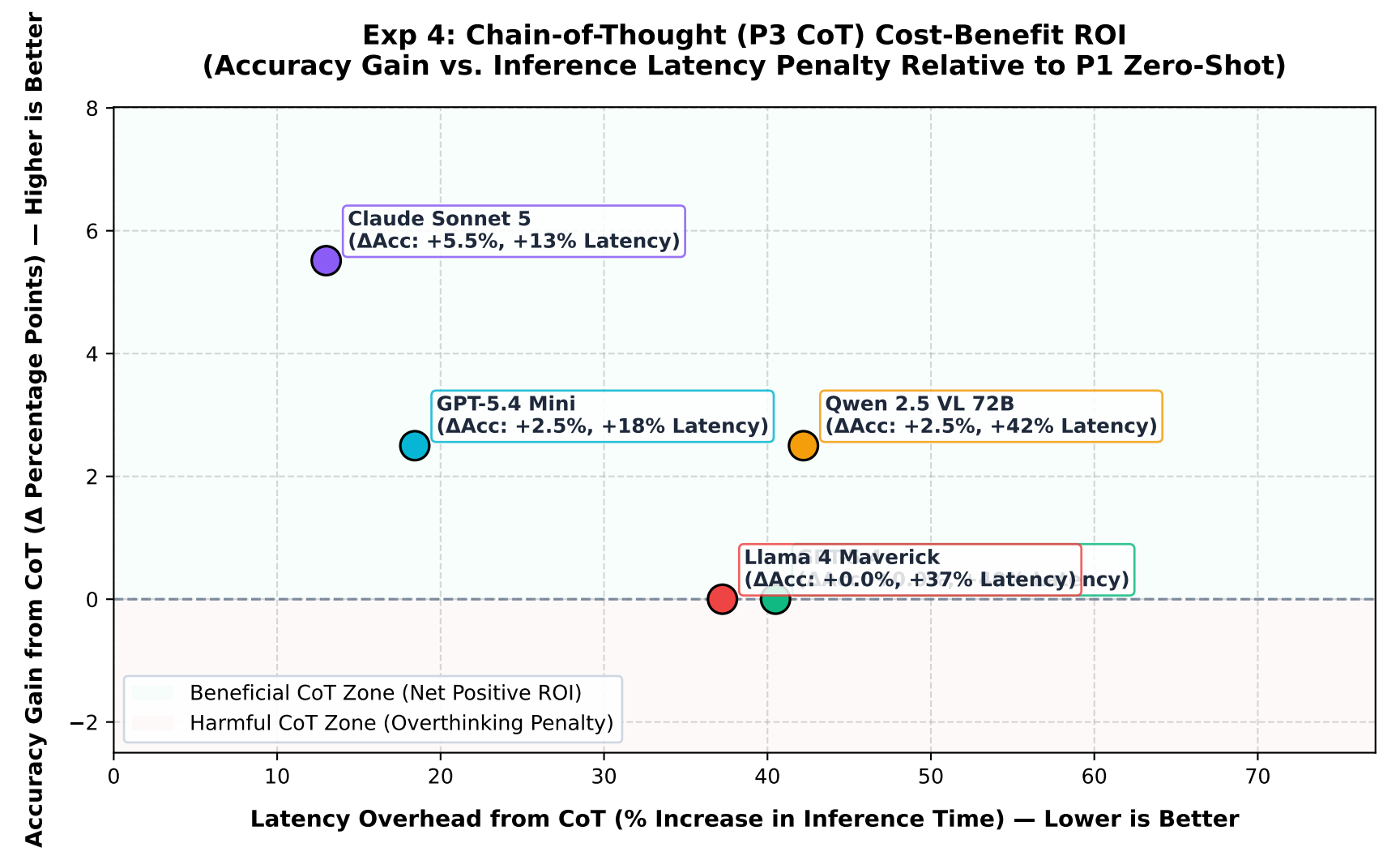
Exp 4: Chain-of-Thought (P3 CoT) Cost-Benefit ROI
(Accuracy Gain vs. Inference Latency Penalty Relative to P1 Zero-Shot)
Accuracy Gain from CoT (Δ Percentage Points) — Higher is Better
Latency Overhead from CoT (% Increase in Inference Time) — Lower is Better
Claude Sonnet 5
(ΔAcc: +5.5%, +13% Latency)
GPT-5.4 Mini
(ΔAcc: +2.5%, +18% Latency)
Qwen 2.5 VL 72B
(ΔAcc: +2.5%, +42% Latency)
Llama 4 Maverick
(ΔAcc: +0.0%, +37% Latency)
Beneficial CoT Zone (Net Positive ROI)
Harmful CoT Zone (Overthinking Penalty)
8
6
4
2
0
−2
0
10
20
30
40
50
60
70